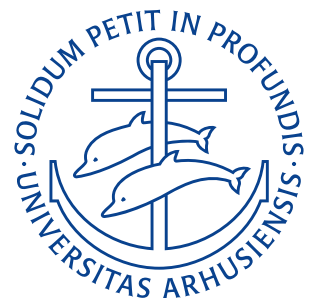


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Homework Exercises

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Date: February 23, 2026



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Lecture 1: Introduction

26. Januar 2026

Problem 1.1a

Show that

$$\delta_{ij}\delta_{ij} = 3.$$

DerivationExpanding $\delta_{ij}\delta_{ij}$ we get

$$\delta_{ij}\delta_{ij} = \delta_{11}\delta_{11} + \delta_{12}\delta_{12} + \delta_{13}\delta_{13} + \delta_{21}\delta_{21} + \delta_{22}\delta_{22} + \delta_{23}\delta_{23} + \delta_{31}\delta_{31} + \delta_{32}\delta_{32} + \delta_{33}\delta_{33}.$$

As $\delta_{11} = \delta_{22} = \delta_{33} = 1$ the above becomes:

$$\delta_{ij}\delta_{ij} = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 3.$$

$\delta_{ij}\delta_{ij} = 3$

RESULT

Problem 1.1c

Show that

$$\alpha_{ij}b_{jk} = \alpha_{in}b_{nk}.$$

Derivation

We expand by summing over the common index j :

$$\alpha_{ij}b_{jk} = \sum_j \alpha_{ij}b_{jk} = \alpha_{i1}b_{1k} + \alpha_{i2}b_{2k} + \alpha_{i3}b_{3k}.$$

Now we sum over n instead:

$$\alpha_{i1}b_{1k} + \alpha_{i2}b_{2k} + \alpha_{i3}b_{3k} = \sum_n \alpha_{in}b_{nk}.$$

Q.E.D.

$\alpha_{ij}b_{jk} = \alpha_{in}b_{nk}$

RESULT

Problem 1.1e

Show that

$$\delta_{ij}C_{jkl} = C_{ikl}.$$

Derivation

Expanding we get

$$\delta_{ij}C_{jkl} = \delta_{11}C_{1kl} + \delta_{22}C_{2kl} + \delta_{33}C_{3kl} = C_{ikl}.$$

$\delta_{ij}C_{jkl} = C_{ikl}$

RESULT

Lecture 2: Traction, Stress, and Equilibrium

2. Februar 2026

Lecture 3: The Cauchy Theorem & Principal Stresses

9. Februar 2026

Problem 2.1

Find the components of the traction on a plane defined by $n_1 = 1/\sqrt{2}$, $n_2 = 1/\sqrt{2}$, and $n_3 = 0$ for the following states of stress:

- (a) $\sigma_{11} = \sigma, \sigma_{12} = 0, \sigma_{13} = 0, \sigma_{21} = \sigma, \sigma_{22} = 0, \sigma_{33} = \sigma.$
 (b) $\sigma_{11} = \sigma, \sigma_{12} = \sigma, \sigma_{13} = 0, \sigma_{22} = \sigma, \sigma_{23} = 0, \sigma_{33} = 0.$

Derivation

- (a) The stress state in (a) is

$$\sigma = \begin{bmatrix} \sigma & 0 & 0 \\ \sigma & 0 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma \end{bmatrix}.$$

The Cauchy Theorem states that

$$T_i = \sigma_{ji} n_j \implies T = \sigma \cdot n.$$

Our plane is:

$$n = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}.$$

Hence,

$$T = \begin{bmatrix} \sigma & 0 & 0 \\ \sigma & 0 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{pmatrix} \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{pmatrix}.$$

- (b) For (b) our stress state is:

$$\sigma = \begin{bmatrix} \sigma & \sigma & 0 \\ \sigma_{21} & \sigma & 0 \\ \sigma_{31} & \sigma_{32} & 0 \end{bmatrix},$$

so

$$T = \begin{bmatrix} \sigma & \sigma & 0 \\ \sigma_{21} & \sigma & 0 \\ \sigma_{31} & \sigma_{32} & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{pmatrix} \sqrt{2}\sigma \\ \frac{\sigma}{\sqrt{2}} + \frac{\sigma_{21}}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{pmatrix}.$$

(a) $T = \begin{bmatrix} \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{bmatrix}$

(b) $T = \begin{bmatrix} \sqrt{2}\sigma \\ \frac{\sigma}{\sqrt{2}} + \frac{\sigma_{21}}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{bmatrix}.$

RESULT

Problem 2.6

At a given point in a body,

$$\sigma_{11} = \sigma_{22} = \sigma_{33} = -p,$$

$$\sigma_{12} = \sigma_{23} = \sigma_{31} = 0.$$

Show that the normal stresses are equal to $-p$ and that the shearing stresses vanish for any other Cartesian coordinate system.

Derivation

Here, our stress tensor is

$$\boldsymbol{\sigma} = \begin{bmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{bmatrix} = -p\mathbf{I}.$$

Using the Cauchy theorem $\mathbf{T} = \boldsymbol{\sigma} \cdot \mathbf{n}$ we get:

$$\mathbf{T} = (-p\mathbf{I}) \cdot \mathbf{n} = -p\mathbf{n}.$$

Hence, $\mathbf{T}$ is parallel to $\mathbf{n}$ meaning the traction does not have any components tangent to the plane and hence the stress is entirely normal with size $-p$.

Lecture 4: Equilibrium Equations & Principle of Angular Momentum

16. Februar 2026

Problem 2.2

The state of stress at a point P in a structure is given by

$$\begin{array}{lll} \sigma_{11} = 20000, & \sigma_{22} = -15000, & \sigma_{33} = 3000 \\ \sigma_{12} = 2000, & \sigma_{23} = 2000, & \sigma_{31} = 1000. \end{array}$$

- Compute the scalar components T_1 , T_2 , and T_3 of the traction $\mathbf{T}$ on the plane passing through P whose outward normal vector $\mathbf{n}$ makes equal angles with the coordinate axes x_1 , x_2 , x_3 .
- Compute the normal and tangential components of stress on this plane.
- Determine the principal stresses.
- Determine the direction of maximum principal stress.

Problem 2.3

The state of stress at a point is given by the components of stress tensor σ_{ij} . A plane is defined by the direction cosines of the normal $(1/2, 1/2, 1/\sqrt{2})$. State the general conditions for which the traction on the plane has the same direction as the x_2 -axis and a magnitude of 1,0.

Problem 2.4

Determine the body forces for which the following stress field describes a state of equilibrium

$$\sigma_{xx} = -2x^2 - 3y^2 - 5z,$$

$$\sigma_{xy} = x + 4xy - 6,$$

$$\sigma_{yy} = -2y^2 + 7,$$

$$\sigma_{xz} = -3x + 2y + 1.$$

$$\sigma_{zz} = 4x + y + 3z - 5,$$

$$\sigma_{yz} = 0.$$

Problem 2.5

Determine whether the following stress field is admissible in an elastic body when body forces are negligible.

$$\boldsymbol{\sigma} = \begin{bmatrix} yz + 4 & z^2 + 2x & 5y + z \\ \cdot & xz + 3y & 8x^3 \\ \cdot & \cdot & 2xyz \end{bmatrix}.$$

Problem 2.8

The stress tensor at a point $P(x, y, z)$ in a solid is

$$\boldsymbol{\sigma} = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}.$$

- Show that the principal stresses are 4, 1, and -2 .
- Find the orientation of the principal planes

Problem 2.9

The stress tensor at a point $P(x, y, z)$ in a solid is

$$\boldsymbol{\sigma} = \begin{bmatrix} 120 & 40 & 30 \\ 40 & 150 & 20 \\ 30 & 20 & 100 \end{bmatrix}.$$

- Determine the stresses with respect to a set of axes x' , y' , and z' for which the matrix of direction cosines is

$$[\alpha_{ij}] = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- Compute the principal stresses using both the x -, y -, and z - and x' -, y' -, and z' -axes as the basis.
- Find the orientation of the principal planes with respect to the x -, y -, and z -axes.

Lecture 5: The Strain Tensor

23. Februar 2026

Problem 3.1

The components of a displacement field are

$$u_x = (x^2 + 20) \cdot 10^{-4},$$

$$u_y = 2yz \cdot 10^{-3},$$

$$u_z = (z^2 - xy) \cdot 10^{-3}.$$

- (a) Consider two points in the undeformed system, $(2, 5, 7)$ and $(3, 8, 9)$. Find the change in distance between these points.
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